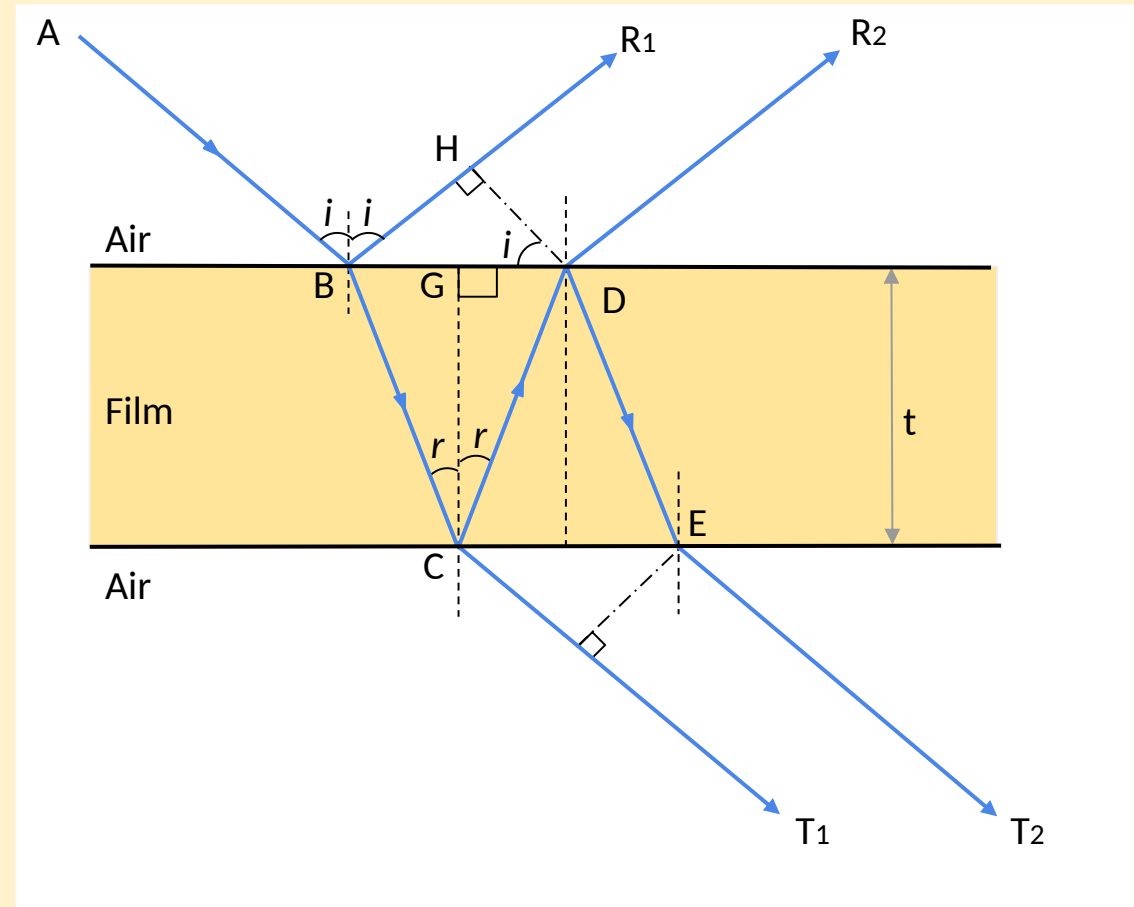


INTERFERENCE

Interference in a thin, transparent film.

- Refraction while going from a *rarer to denser* medium generates an additional *phase difference of π* , which results in additional *path difference of $\lambda/2$* .
- For a *reflected system*:
 1. Condition for *maxima*: $2\mu t(\cos r) = (2n-1)(\lambda/2)$
 2. Condition for *minima*: $2\mu t(\cos r) = n\lambda$
- For a transmitted system:
 1. Condition for *maxima*: $2\mu t(\cos r) = n\lambda$
 2. Condition for *minima*: $2\mu t(\cos r) = (2n-1)(\lambda/2)$
- A *reflected* system has *higher contrast* in the *intensities of the maxima and minima*, while a *transmitted* system has *poor contrast*.

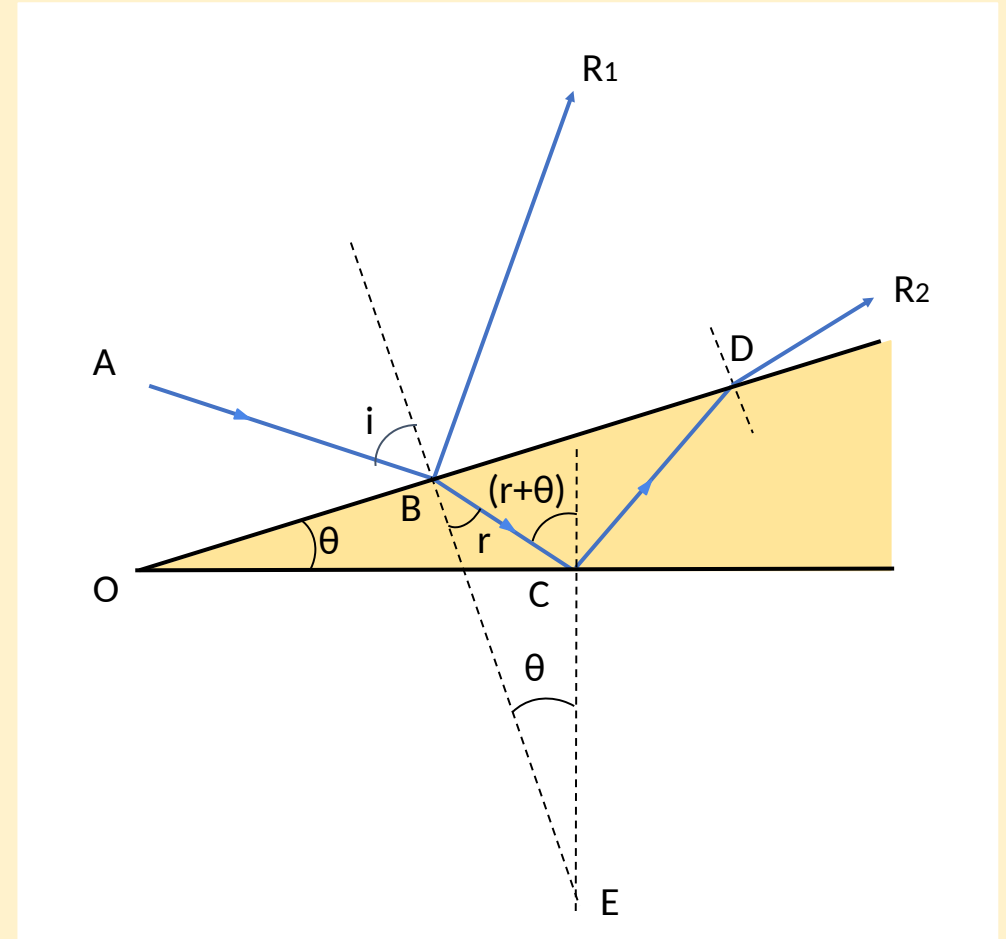
7 times for 4-8 Marks



Wedge-shaped film.

5 times for 3-7 Marks

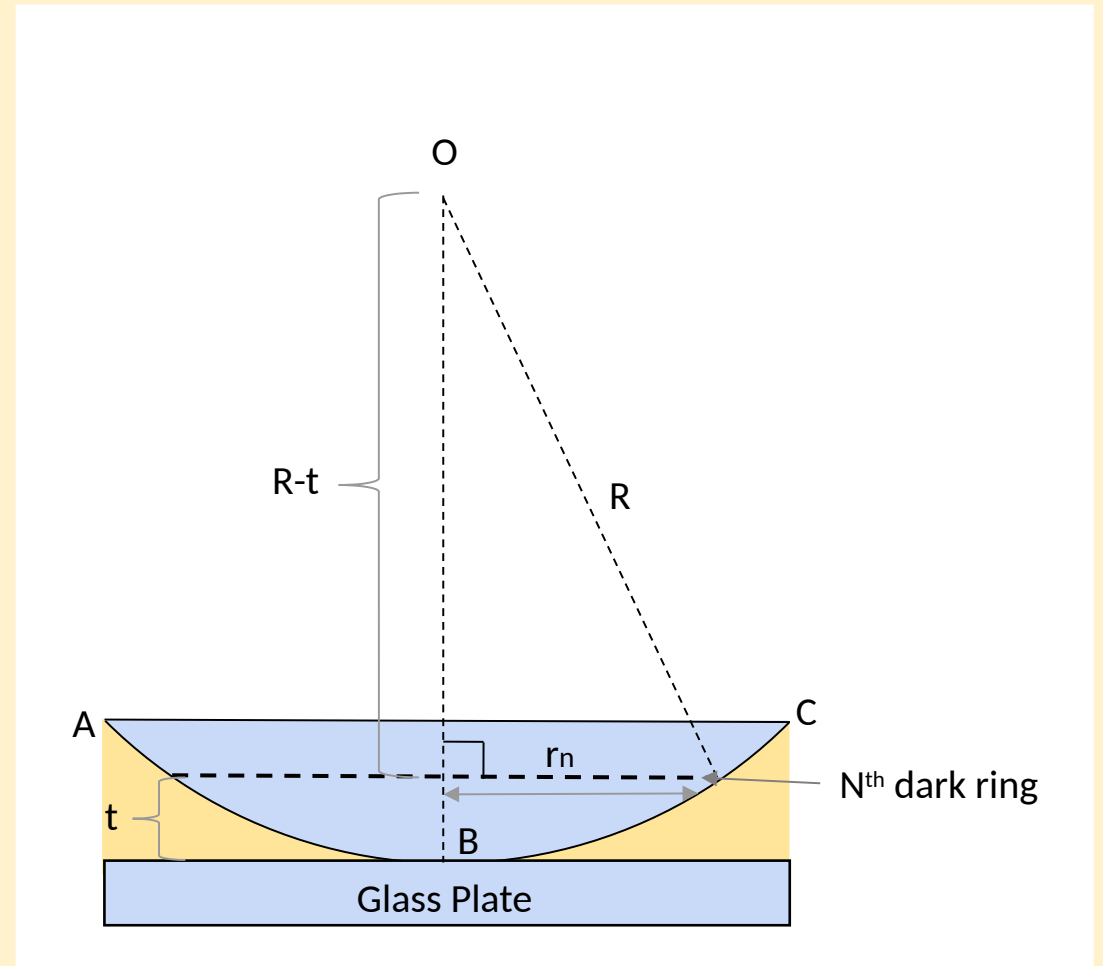
- Condition for constructive interference(*maxima*):
$$2\mu t(\cos (r+\theta)) = (2n-1)(\lambda/2)$$
where $n = 1,2,3,4\dots$
- Condition for destructive interference(*minima*):
$$2\mu t(\cos (r+\theta)) = n\lambda$$
Where $n = 0,1,2,3\dots$



Newton's Rings

- Newton's Rings (NR) setup consist of the plano-convex lens placed on a glass plate.
- Once the setup is illuminated alternate bright and dark rings are formed due to amplitude division type of interference in the thin film that is created between the plano-convex lens and glass plate.
- For a dark ring: $\Delta = (2n+1)(\lambda/2)$
- For a bright ring:
 - $\Delta = n\lambda$
 - $r_n = ((2n-1)(\lambda/2)R)^{1/2}$
 - Thus $r_n \propto (2n-1)^{1/2}$

5 times for 3-7 Marks



Circular Fringes in Newton's Rings/ Straight Fringes in Wedge-shaped films.

4 times for 3 Marks

- Fringes are the *locus* of points of *equal thickness*.
- In Newton's Rings, the points of equal thickness lie on a *circular path (in concentric circles)*.
- In wedge-shaped films, points of equal thickness are *straight, parallel lines*.
- Thus, fringes in Newton's Rings are circular and concentric, while fringes in wedge-shaped films are straight and parallel.

Why is the centre always dark, in Newton's Rings?

1 time for 3 Marks

- Condition for bright rings: $2\mu t(\cos(r+\theta)) = (2n+1)(\lambda/2)$
- Condition for dark rings: $2\mu t(\cos(r+\theta)) = n\lambda$
- Thickness at point of contact $t = 0$
- For large lens radius, $\theta = 0$, and for normal incidence, $r = 0$
- **New** condition for bright rings: $2\mu t = (2n+1)(\lambda/2)$
- **New** condition for dark rings: $2\mu t = n\lambda$
- $t = 0$ only *satisfies* the new *condition for dark rings*, hence the centre is always dark.

Liquid introduced in Newton's Rings

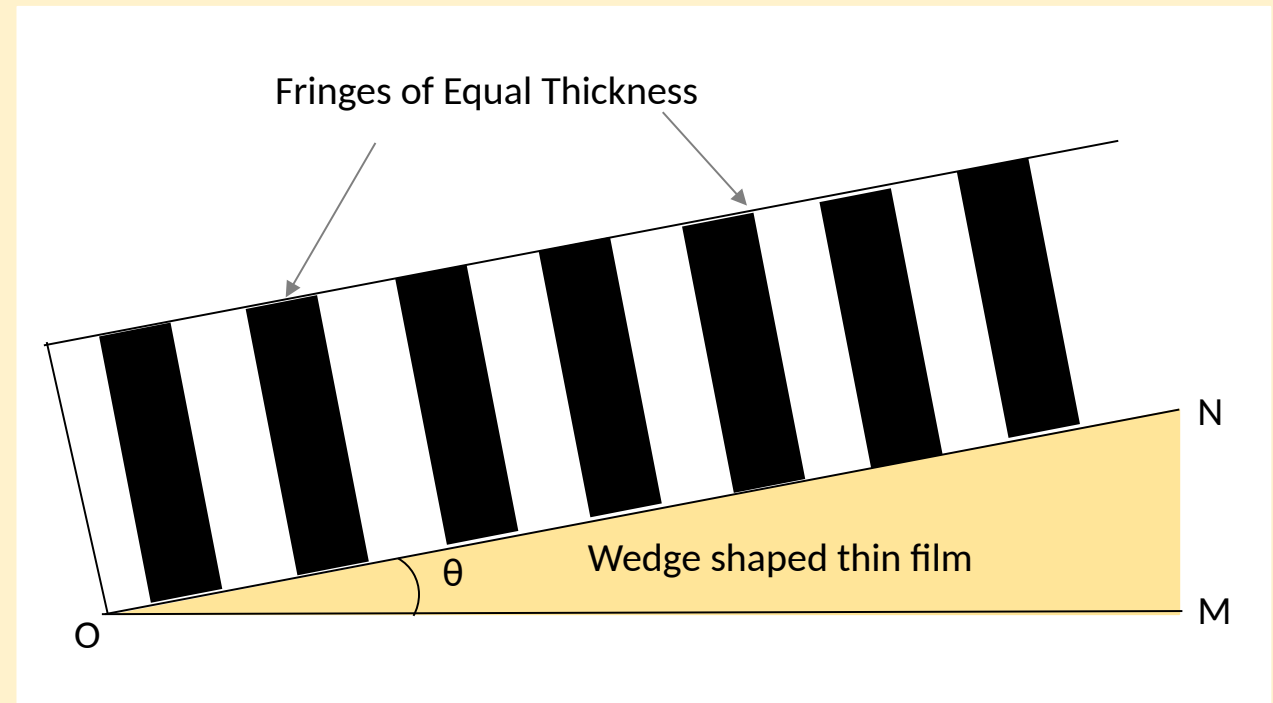
- The *center* of the Newton's Rings experiment, when performed under *normal conditions (air)*, is always *dark*.
- If a liquid, whose μ is *greater than the lens* but *smaller than plate*, is used, the *centre* becomes *bright*.
- This is because light is going from a *rarer to denser medium twice*, which results in a *phase change of π* twice.
- This collectively results in *no phase change* at all.
- Hence, constructive interference occurs which gives rise to a bright centre spot.

Applications of Diffraction

Testing Optical Flatness

1 time for 3 Marks

- Based on *Interference*.
- If ON and OM are flat, they form a *wedge-shaped film interference* system, which varies consistently in thickness.
- *Fringe pattern* is tested.
- If the fringes have *equal thickness*, then the test surface is optically flat.



Applications of Diffraction

Origin of colour in thin films

4 times for 3 Marks

- When white light is incident on a film, *different colours travel different lengths of path* in the film.
- Thus, *different colours are reflected from different points* on the film.
- Some *colour interferes constructively* and is *intensified*, while the *rest fade out*.
- This is how colour originates in thin films.

Applications of Diffraction

Anti-Reflecting Films

4 times for 3-8 Marks

- Light is *lost due to reflection* at glass-air interfaces.
- *Hinders* working of *complicated optical* devices.
- An anti-reflecting film *prevents* this.
- The refractive index of the film is *greater than that of air* but *less than that of the material* it is coated on.

